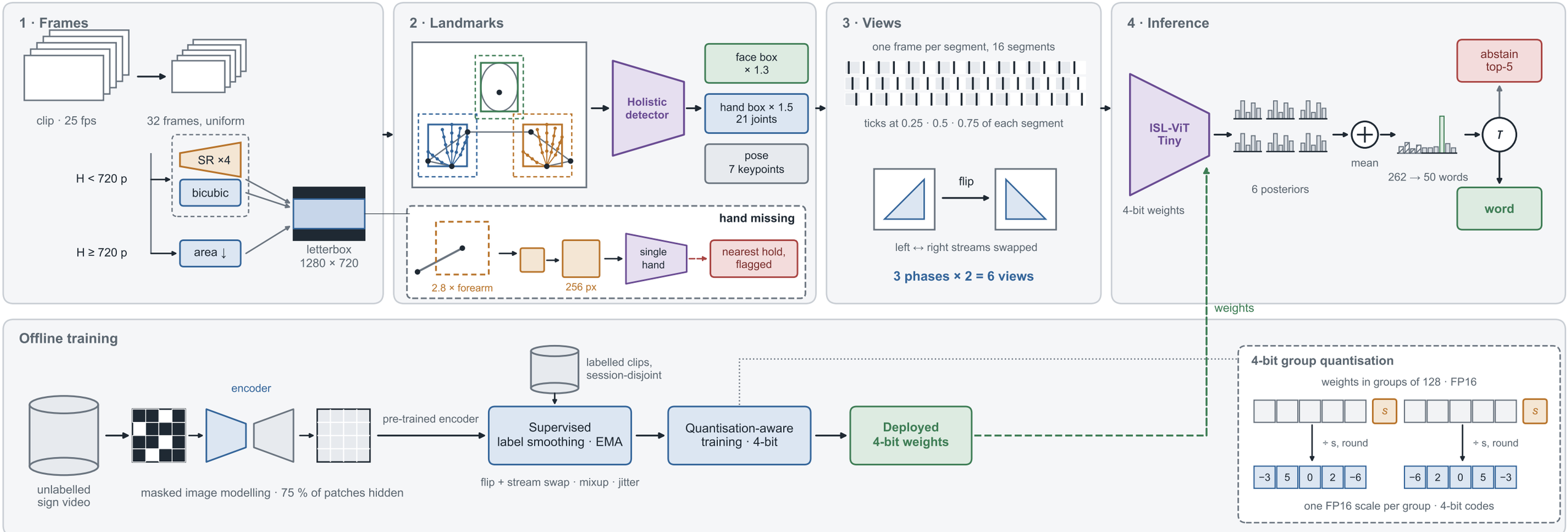
1 · Frames
clip · 25 fps
32 frames, uniform
H < 720 p
SR ×4
bicubic
H ≥ 720 p
area ↓
letterbox
1280 × 720
2 · Landmarks
Holistic detector
face box
× 1.3
hand box × 1.5
21 joints
pose
7 keypoints
hand missing
2.8 × forearm
256 px
single hand
nearest hold, flagged
3 · Views
one frame per segment, 16 segments
ticks at 0.25 · 0.5 · 0.75 of each segment
flip
left ↔ right streams swapped
3 phases × 2 = 6 views
4 · Inference
ISL-ViT Tiny
4-bit weights
6 posteriors
mean
262 → 50 words
T
abstain
top-5
word
weights
Offline training
unlabelled
sign video
encoder
masked image modelling · 75 % of patches hidden
pre-trained encoder
labelled clips,
session-disjoint
Supervised
label smoothing · EMA
flip + stream swap · mixup · jitter
Quantisation-aware
training · 4-bit
Deployed
4-bit weights
4-bit group quantisation
weights in groups of 128 · FP16
s
s
÷ s, round
÷ s, round
−3 5 0 2 −6
−6 2 0 5 −3
one FP16 scale per group · 4-bit codes